

Mid-Term Project Guidelines

IMPORTANT ACTION REQUIRED: Group Registration

Whether you are working individually or in a team, **you must register your group status** via the following link:

 [Group Registration Form \(Tencent Docs\)](#)

Project Overview

For the mid-term project, students can work individually or form teams (*Note: Please ensure you have registered your choice using the link above*). You are required to submit a Report and deliver a PPT Presentation. You must choose one of the following two tracks:

- **Option A (Research-oriented):** Select one paper from our reading list (which may also be your presentation paper), read it closely, and then explore the papers that cite it in order to identify a research stream for a literature review. This review should both provide a foundation for your own research and offer a critical analysis of the existing literature. Your literature review must evaluate at least **one core paper** and at least **three additional related papers**.
- **Option B (Application-oriented):** Identify a high-quality open-source GitHub repository related to Large Language Models (LLMs), and analyze its current capabilities, limitations, and potential for improvement. Based on your analysis, propose either a concrete optimization plan or the design of a new feature. No code submission is required for the mid-term; instead, focus on conceptual design and the strength of your proposal. Your review should evaluate at least **one core GitHub repository** and refer to at least **three additional related repositories**.

1. Mid-Term Presentation Guidelines (PPT)

- **Time:** 15 minutes plus 5 minutes Q&A per group.
- **Format:** PowerPoint.
- **Presentation Order:** To be announced in class.
- **Presentation Mode:** Students who are not scheduled to present live will submit a recorded presentation.

Suggested Slide Structure

1. **Motivation & Importance (3 min)**
 - Briefly introduce the problem (Option A) or the chosen repository (Option B) and why it is important.
 - Provide real-world or research context.
 2. **Literature Review / Repository Analysis (4 min)**
 - Summarize what has already been done in this area or the current status of the repository.
 - Point out limitations in existing methods, models, or the current codebase.
 3. **Your Expected Contribution (3 min)**
 - Clearly state how your proposed approach or feature is different or better.
 - Emphasize novelty or practical relevance.
 4. **Proposed Methodology & Action Plan (4 min)**
 - Explain your planned method or conceptual design for code improvement concisely.
 - Include visual aids if applicable (e.g., diagrams, flowcharts, architecture plans).
 5. **Conclusion & Next Steps (1 min)**
 - Wrap up with takeaways.
 - Mention any anticipated challenges and propose your timeline/directions for the final project.
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2. Mid-Term Report Guidelines

- **Format:** PDF file only.
- **Submission:** Submit on Blackboard (BB).

Report Structure

Please organize your report with the following sections:

1. **Introduction** * Clearly state the problem you are exploring or the repository you are analyzing.
 - Explain why this problem matters (societal, theoretical, or practical importance).
 - Summarize existing literature or the current state of the repository.
 - Highlight gaps or limitations in the current literature or codebase.
 - Clearly state your planned innovation, contribution, or optimization goal.

2. **Proposed Methodology & Design** * Describe the approach you plan to use to solve the problem or optimize the repository.
 - Include any models, algorithms, frameworks, or data sources you intend to use.
 - Justify your design choices where appropriate.
 3. **Expected Outcomes & Discussion**
 - Reflect on the potential applications of your proposed work.
 - Discuss the potential limitations or challenges of your planned approach.
 - Outline your concrete next steps to implement this plan before the final deadline.
 4. **References** * Include any references cited in your work using a consistent citation format.
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3. Mid-Term Evaluation Rubric

(Focuses on planning, problem identification, and feasibility)

Category	Description
1. Contribution & Innovation	• Is the project tackling a meaningful problem or proposing a valuable repository extension?
	• Does the proposed approach offer a novel solution or practical improvement?
	• Does the student clearly articulate what will be new compared to existing work?
2. Methodological Feasibility	• Is the proposed method or optimization plan technically sound and logically appropriate?
	• Are assumptions clearly stated?
	• Are the planned tools, data, or algorithms feasible?
3. Analysis & Evaluation	• Are the literature review or repository analyses presented clearly and interpreted thoughtfully?
	• Are claims supported by evidence or careful code analysis?
	• Are current limitations and potential future challenges discussed honestly?

